[INCH-POUND]

MIL-C-0083522/2E(NASA)
1 JANUARY 1989
USED IN LIEU OF
MIL-C-83522/2D

MILITARY SPECIFICATION SHEET

CONNECTOR, FIBER OPTIC, HIGH RELIABILITY, FIXED SINGLE TERMINUS, PLUG 100/140 µm FIBER (SPACE LEVEL)

This specification is approved for use by the Aeronautics and Space Administration based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-83522/2D. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83522.

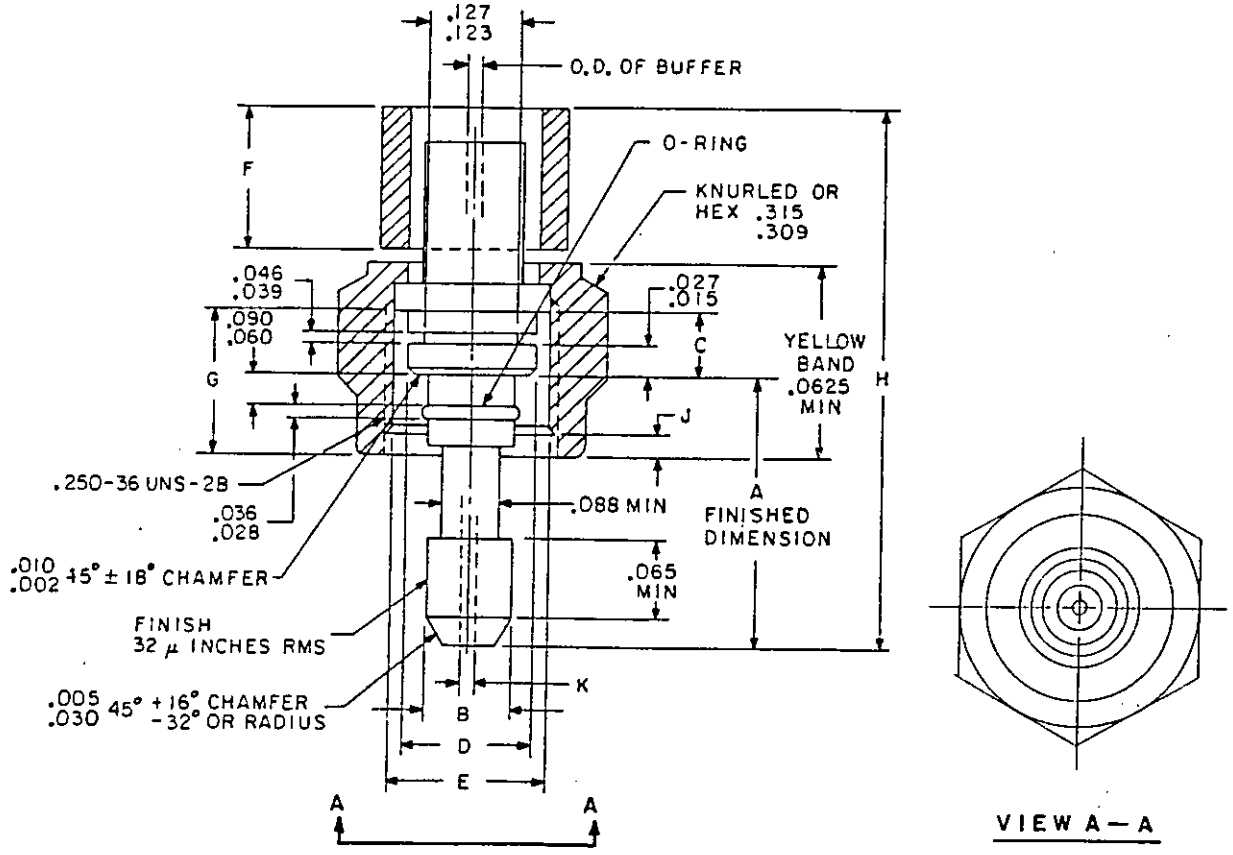


FIGURE 1. Dimensions and configuration.

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	Min	Max
A 4/	.3858 (9.799)	.3863 (9.812)
C	.080 (2.03)	.100 (2.543)
D	.179 (4.55)	.182 (4.62)
E	.220 (5.59)	.233 (5.92)
F 5/	.485 (12.3)	
G	.183 (4.65)	.224 (5.69)
H		1.50 (38.1)
J	.015 (0.38)	.045 (1.14)
B	Plug must not enter .118 hole gauge	Plug must enter .1250 dia. hole gauge with a max force of 12 lbs.

K dimension
100/140 µm fiber
145.0 max 142.5 min

Inches	mm
.002	0.05
.005	0.13
.010	0.25
.015	0.38
.027	0.69
.028	0.71
.030	0.76
.036	0.91
.039	0.99
.046	1.17
.060	1.52
.0625	1.588
.065	1.65
.088	2.24
.090	2.29
.250	6.35
.309	7.85
.315	8.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses (K dimension is in µm).
4. Dimensions A is obtained after the connector has been assembled.
5. Ferrule may be an integral part of the connector.
6. Cable dimensions shall be in accordance with MIL-C-85045/12.
7. Three holes .016 (0.41 mm) in diameter, equally spaced on the coupling nut, are required for safety wiring after mating.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Metals: Plug body and coupling nut shall be corrosion resistant (stainless) steel in accordance with QQ-S-763 class 303, nonmagnetic type or nickel plated brass in accordance with QQ-B-626.

Materials: Connector tip shall be of polyetherimide. No other materials (such as silicone lubricants) shall be used in processing.

Epoxies: Use Trabond 230, Eccobond 144B or an equivalent epoxy approved by the qualifying activity.

Dimensions and configuration: See figure 1.

Fiber optic cable requirements:

Cable diameter: In accordance with DOD-C-85045.

Cable configuration: In accordance with DOD-C-85045.

Fiber diameter: 100/140 µm.

Fiber numerical aperture: In accordance with DOD-C-85045.

Fiber type: In accordance with DOD-C-85045.

Fiber optic contact:

Method of optical alignment: Precision diameter fit.

Lens configuration: Not applicable.

Coating requirements: Not applicable.

Optical requirements:

Number of optical termini: One.

Coupling loss (attenuation): 2 dB maximum in accordance with mated pair.

Weight: 6 grams maximum.

Safety wire holes: Required.

Force to engage and disengage:

Longitudinal force: 12 pounds maximum.

Torque: 4 inch-pounds maximum.

Coupling proof torque: 5 inch-pounds minimum.

Coupling mechanism retention force: 60 pounds minimum.

Mating durability: Connectors shall be tested as specified in EIA-455-21. Connector attenuation shall not increase by more than 1 dB after 200 mate and demate cycles. Visual examination shall be performed after the test.

Pressure altitude: The pressure shall be 1 x 10^{-6} atm, and temperature shall be at the maximum rated operating value. Connector attenuation increase shall not exceed 1 dB at any interval of the test.

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Temperature life: Connectors shall be tested as specified in EIA-455-4, test conditions 5 and F. Connector attenuation shall not increase by more than 1 dB at any interval of life test.

Submersion: Not applicable.

Salt spray: Not applicable.

Dust: Not applicable.

Ozone exposure: Not applicable.

Fluid immersion: Not applicable.

Material outgassing: Connector and assembly materials shall not exceed the limits of 1 percent total mass loss (TML) and 0.1 percent collected volatile condensable materials (CVCM) when tested as specified in ASTM-E-595.

Qualification inspection shall consist of performing the inspections and optical tests specified in table I herein, in the sequence shown.

Quality conformance inspection shall be performed as specified in MIL-C-83522 and as specified herein.

Group A inspection shall consist of performing the inspections and optical tests specified in table II herein, in the sequence shown on all parts.

Group B inspection shall consist of performing the inspection and optical tests specified in table III herein, in the sequence shown.

TABLE I. Qualification inspection.

Inspection	Sample quantity
Group I 1/	20 units
Weight	20 units
Size	20 units
Coupling loss	20 units
Group II	4 units
Temperature	4 units
Humidity	4 units
Marking	4 units
Group III	4 units
Tensile loading	4 units
Cable strain relief	4 units
Flex life	4 units
Twist	4 units
Force to engage/disengage	4 units
Coupling proof torque	4 units
Coupling mechanism retention force	4 units

See footnote at end of table.

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TABLE I. Qualification inspection - Continued.

Inspection	Sample quantity
Group III - continued	4 units
Mating durability	4 units
Temperature cycling	4 units
Vibration	4 units
Shock	4 units
Marking	4 units
Group IV	4 units
Mating durability	4 units
Pressure altitude	4 units
Temperature cycling	4 units
Humidity	4 units
Marking	4 units
Group V	8 units
Materials outgassing	4 units
Temperature life	4 units

1/ All test units shall have completed group A tests in accordance with table II. Group II through group V samples shall be taken from group I.

TABLE II. Group A inspection (100%).

Inspection
Visual and mechanical
Weight
Size
Workmanship

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TABLE III. Group B inspection.

Inspection	Sample quantity
Group I 1/	4 units
Coupling loss	4 units
Group II	4 units
Temperature cycling	4 units
Temperature	4 units
Vibration	4 units
Tensile load	4 units
Group III	4 units
Cable strain relief	4 units
Force to engage/disengage	4 units
Coupling proof torque	4 units
Coupling mechanism retention force	4 units
Marking	4 units

1/ Group I samples shall have completed group A tests.

Part or identifying number (PIN): M83522/2-01S.

Plug accessories: Each plug shall be supplied with a one crimp ferrule as needed (see figure 1); one protective cable-end cap; one 1.5 inch length of heat-shrink tubing or one molded thermoplastic strain relief and assembly instructions. The assembly instructions shall include a listing of the epoxies which meet outgassing requirements of this specification sheet.

Cleaning procedures: Dampen lens tissue (paper wipes or cotton swab) with a small amount of pure isopropyl alcohol or deionized water. Gently wipe the face of the connector, removing any debris, particularly around the optical fiber, using clean lens tissue (or any lint free wipes). Carefully dry contact with clean, dry, and lint free tissue.

NOTE: The plug mates with MIL-C-83522/4 receptacle; MIL-C-83522/5 receptacle; MIL-C-83522/7 receptacle; MIL-C-83522/8 receptacle; or a MIL-C-83522/3 adapter mated with another MIL-C-83522/2 plug.

Applicable cables shall be specified in accordance with MIL-C-85045/12.

CONCLUDING MATERIAL

Custodian:
NASA - NA

Review activity:
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 6060-K023)

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